

Ayush Saha

[Github](#) [LinkedIn](#) [Portfolio](#)

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EDUCATION

Siliguri Institute of Technology

Bachelor of Engineering - Computer Science and Engineering

West Bengal

2023 - 2027

TECHNICAL SKILLS

Languages/Technologies: Java, Python, C

Databases/Tools: PostgreSQL, MySQL, Redis, Qdrant, Supabase, Neon, Git, Docker

Libraries/Frameworks: LangChain, LangGraph, LlamaIndex, FastMCP, FastAPI

Generative AI: Natural Language Processing, Transformers, Open Source LLMs, Retrieval Augmented Generation, AI Agents, Model Context Protocol

PROJECTS

RAG powered Enterprise Document Assistant | *Python, LangChain, Qdrant, FastAPI, Redis* [Demo](#) | [Github](#)

- Engineered **IntelliDocs**, a scalable RAG platform supporting **100k+ documents** with a sustained throughput of **60-120 queries/min** using **Python, FastAPI**, and **Supabase**.
- Designed a sophisticated **multi-agent pipeline** (Planner → Router → Retriever → Synthesizer → Validator) that improved **answer reliability** by **~35%** and **citation precision** by **~25%** through automated query classification and timeout guardrails.
- Optimized document processing to achieve **60+ docs/sec** (a **6-8x speedup**) by implementing **threaded loading**, **process-based chunking**, and **batched embedding generation**.
- Reduced **LLM API spend** by **50%** and **median latency** by **55%** by building a **four-level caching layer** (KV, retrieval, LLM response, and human-approved) with strict **session-scoped isolation**.

RAG Accuracy Evaluating MCP Tool | *Python, FastMCP, FastAPI, Neon, Redis*

[Github](#)

- Built an installable **RAG accuracy evaluation platform** with **15 production-facing tools** and a **Postgres-backed job flow**, reducing RAG evaluation setup time by **80%**.
- Implemented **deterministic run caching** and **metric fingerprinting** across **8 evaluation metrics**, cutting redundant re-runs by **65%** and lowering repeat LLM/API spend by an estimated **40%**.
- Architected an **asynchronous API-plus-worker pipeline** with a **1.0s polling interval**, reducing processing latency by **75%** and enabling sustained throughput of **1 eval run/sec per worker**.
- Engineered **automated failure clustering** and **Gemini-based suggestions** to group issues into the **top 5 clusters**, reducing root-cause triage time by **60%**.
- Standardized experiment reliability through **MCP contracts** and **API-key isolation**, eliminating **100% of cross-project data leaks** and increasing regression-detection coverage across **12 automated flows**.

Agent Evaluated Code Test Generator | *Python, LangGraph, Redis, FastAPI, Neon*

[Demo](#) | [Github](#)

- Engineered a high-performance AI test generation platform using **FastAPI** and **Redis-backed multi-layer caching**, achieving a **55% reduction in latency** via AST-aware parser reuse.
- Architected an **LLM orchestration pipeline** using **LangGraph** for automated code analysis and self-correction, increasing first-pass syntactic validity to **85%** across multi-language workflows.
- Optimized system reliability by implementing **PostgreSQL job persistence** and idempotent request handling, resulting in **90%+ fewer duplicate generation jobs**.
- Accelerated developer feedback loops by integrating **PostgreSQL-based artifact storage**, improving end-to-end usability and traceability for AI-assisted QA tasks.